

Yang-Shuo Hsiung

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PhD Student
Leung Center for Cosmology and Particle
Astrophysics (LeCosPA)
Max Planck-IAS-NTU Center for Particle Physics,
Cosmology and Geometry
National Taiwan University (NTU)

Education

PhD, Physics, National Taiwan University	08/2026 – Present
MS <i>with honors</i> , Physics, National Taiwan University	08/2023 – 07/2026
BS, Physics, National Taiwan University	07/2019 – 01/2023
Gap-Year, Physics, Tsinghua University	08/2018 – 06/2019

Research Experiences

Summer Student, IHES Summer School – Cosmological Correlators	07/2026 – 07/2026
Visiting Student Researcher, EFI, University of Chicago	06/2025 – 09/2025
Visiting Student Researcher, OUTAP, Osaka University	07/2024 – 08/2024

Publications

Y.-S. Hsiung & P. Chen, <i>A New Approach to the Calculation of Particle Creation from Analog Black Holes</i> , arXiv:2511.22895 [gr-qc] (2025).	11/2025
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Honors, Grants, and Certifications

• Honorary Membership , The Phi Tau Phi Scholastic Honor Society of the Republic of China	05/2026
• Dean's List Award of Graduation , College of Sciences, National Taiwan University	05/2026
• E.SUN Outstanding Talent Scholarship , E.SUN Volunteer & Social Welfare Foundation	10/2025
• Chee-Chun Leung Scholarship , Department of Physics, National Taiwan University	10/2025
• Certificate of Completion , UCTS program, University of Chicago	08/2025
• UCTS Fellowship , UCTS program, University of Chicago	02/2025
• Certificate of Achievement , NASA's Open Science 101 Curriculum, NASA Transform to Open Science	10/2024
• CoS Travel Grants and Scholarship , College of Science, National Taiwan University	09/2024
• Certificate of Completion , ISP Program, Graduate School of Science/ School of Science, Osaka University	08/2024
• Certificate of Completion , Academic Research Ethics Education Courses, Center for Taiwan Academic Research Ethics Education	08/2024
• Osaka University Graduate School of Science Scholarship , ISP Program, Graduate School of Science/ School of Science, Osaka University	07/2024
• JASSO Scholarship , ISP Program, Japan Student Services Organization	07/2024
• Dean's List Award , National Taiwan University	01/2021
• Outstanding Student Representative , Tsinghua University	10/2018
• Full Score (PR 99.8) in 2018-GSAT , Ministry of Education, Taiwan	01/2018
• 3rd Place in Taiwan Young-Student Physicists' Tournament , Far Eastern Y. Z. Hsu Foundation, Taiwan	03/2017

Research Projects

<i>A New Approach to the Calculation of Particle Creation from Analog Black Holes</i> Advisor: P. Chen (LeCosPA, NTU) & R. M. Wald (EFI, UChicago)	07/2023 – Present
<i>Simulation Approach to the Entanglement Study of Analog Black Holes</i> Advisor: P. Chen (LeCosPA, NTU)	09/2024 – 11/2025
<i>Investigation into Cosmological Correlation and Baryonic Feedback Effects</i> Advisor: K. Nagamine (OUTAP, Osaka Univ.)	07/2024 – 08/2025
<i>The Weak Parametric Resonance Model and the Cosmological Constant Problem</i> Advisor: P. Chen (LeCosPA, NTU)	09/2022 – 01/2023
<i>A Study on Entanglement Entropy and Black Hole Information Loss Paradox</i> Term Project, Advisor: P. Chen (LeCosPA, NTU)	09/2022 – 01/2023
<i>A Test on the Extremal Black Hole and the Cosmic Censorship</i> Term Project, Advisor: Y. Huang (NTU)	02/2022 – 06/2022
<i>Delensing the CMB B-mode via the Cosmic Infrared Background Approach</i> Term Project, Advisor: T.-W. Lan (NTU, ASIAA)	09/2021 – 01/2022
<i>Bootstrapping the Quantum Harmonic Oscillator</i> Term Project, Advisor: Y. Huang (NTU)	09/2021 – 01/2022

Presentations

- [Talk] *A New Approach to the Calculation of Particle Creation from Analog Black Holes*, 05/2026
Master Thesis Oral Examination, NTU
- [Talk] *A New Perspective on Particle Creation from Analog Black Holes*, UCTS Program, 08/2025
UChicago
- [Poster] *A New Perspective on Particle Creation from Analog Black Holes*, UCTS Program, 08/2025
UChicago
- [Talk] *Investigation into the Baryonic Feedback Effect to the Large-Scale Structure Formation*, 08/2024
ISP Program, Graduate School of Science, Osaka Univ.
- [Talk] *Numerical Approach of the Hawking Radiation Spectrum in AnaBHEL Experiment*, 07/2024
OUTAP Seminar, Osaka Univ.
- [Talk] *Learning to Count the Hawking Radiation in AnaBHEL Experiment*, Taipei Astronomy 01/2024
Workshop, ASIAA
- [Lecture] *The Hot Big Bang*, Cosmology Study Group, LeCosPA 12/2023

Skills

Programming: Python, Mathematica, Linux, Bash, LaTeX, Cosmological Codes

Techniques: Numerical Calculation, Computer Simulation, Machine Learning (Coursera)

Leadership and Teaching

Teaching Assistant , General Physics (EMI), NTU	09/2024 – 06/2025
Instructor , Physics Project for Science Fair, Junior High School Student	09/2024 – 01/2025
Competition Leader , Badminton Team of the Department of Physics, NTU	10/2019 – 12/2024
Co-Founder , Cosmology Study Club, LeCosPA, NTU	10/2023 – 06/2024
Teaching Assistant , General Physics, NTU	09/2023 – 06/2024
Instructor , Physics Project for Science Fair, Senior High School Student	07/2023 – 01/2024

Instructor , Math Project for Science Fair, Senior High School Student	07/2023 – 01/2024
Teaching Assistant , Classical and Quantum Black Holes, NTU	09/2023 – 12/2023
Instructor , Team of National Experimental High School at Hsinchu Science Park, Taiwan Young-Student Physicists' Tournament	02/2020 – 03/2020

Outreach

SMART Summer Program , Docent of UChicago, Senior High School Students	07/2025
<i>Unveiling the Mysterious Universe</i> , Talk, NTU	12/2024
<i>A Zoom-In Journey: From Taiwan to LeCosPA</i> , Talk, Osaka Univ.	08/2024
Improvement of Comprehensive Assessment Program for Junior High School Students , Advisor, Junior High School Teachers	08/2023
KISS Science Educational Event , Docent in LeCosPA, Public	10/2022
<i>Learning, Abilities and Education</i> , Talk, Junior High School Students	11/2021
<i>Studying Experiences in Tsinghua University</i> , Talk, Senior High School Students	02/2019